SHINDE VINAYAK RAO PATIL

AI / ML Engineer · NLP & Speech Processing · Real-Time AI Systems

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PROFESSIONAL SUMMARY

Master of Science graduate in Data Science from IIIT Hyderabad specialising in NLP, deep learning, and real-time speech AI. Currently building next-generation bilingual Arabic–English voice AI systems at Nabhe, with hands-on experience across the full ML lifecycle - from data preparation and model fine-tuning to production deployment on specialised hardware (Qualcomm Cloud AI 100). Holds a published journal paper, an accepted conference paper, and a filed patent, alongside recognition as a top hackathon finisher and departmental award.

PROFESSIONAL EXPERIENCE

AI Engineer · Nabhe, Subsidiary of Master Works *Jan 2026 – Present | Hyderabad*

Real-Time Speech Translation Platform - MOFA (Ministry of Foreign Affairs)

- Built a real-time speech translation platform (STT + LLM-based machine translation) supporting a 9-language pipeline (Arabic, English, Spanish, Japanese, Korean, Mandarin, Tagalog, Portuguese, French), with diplomatic terminology enforcement, low-confidence/missing-word detection, automatic session archiving, and administrative reporting.
- Integrated LLM-based translation via Groq (openai/gpt-oss-120b) using streaming WebSocket inference with vLLM, achieving ~300ms end-to-end latency (STT→MT) in production benchmarks.
- Designed a fully decoupled, configuration-led architecture with per-language/per-dialect model routing, directing each supported dialect to its specialized ASR backend.
- Designed and deployed the end-to-end application on-premises on Qualcomm Cloud AI 100 (QAIC) hardware, running both STT and LLM translation models directly on Qualcomm NPU infrastructure.

AI Automation Intern · Talbot Health Services (via DSIG IT Solutions) *Sep 2025 – Dec 2025 | Remote (US-based, parallel)*

- Built a RAG-based system to automate internal SQL query and operations retrieval for a US healthcare client; reduced manual lookup time significantly.
- Developed agentic AI workflows for denial management and automated rules-engine orchestration using Python, voice transcription, and Stripe integration.

AI/ML Intern · BYOL Academy *Sep 2025 – Dec 2025 | Remote (parallel)*

- Built a production-grade, AI-powered RAG chatbot for medical preparation, including knowledge base construction, error correction for current affairs content, and multilingual translation support.

NLP Research Intern · Speech Processing Lab, IIIT Hyderabad *Jul 2023 – Nov 2023 | Hyderabad*

- Researched emphasis patterns in spoken language vs. text summarisation using TEDx data; built keyword extraction and summary generation models for spoken content, contributing to advancements in NLP.

AI Developer Intern · Plausibility Solutions *Oct 2023 – Apr 2024 | Bengaluru*

- Built and deployed multiple RAG-based chatbots using LLMs; designed end-to-end RAG pipelines for contextual response generation with Pinecone vector store.
- Fine-tuned LLMs for domain-specific retrieval accuracy; developed multi-agent orchestration systems using Agentic AI.

AI Developer Intern · Eunoia Innovations Pvt. Ltd. *Feb 2022 – Feb 2023 | Hyderabad*

- Analysed water sensor data and integrated ML/IoT inference with hardware components (thrusters, actuators).

EDUCATION

M.S. in Data Science (CGPA:7.8) *2024 – 2026*

International Institute of Information Technology Hyderabad (IIIT-H)

B.Tech. in Computer Science and Engineering (CGPA: 7.00) *2020 –2024*

Hyderabad Institute of Technology and Management (HITAM)

PERSONAL PROJECTS

TASA - Real-Time Streaming Voice-to-Voice AI Assistant

- Built a fully local voice-to-voice AI (faster-whisper STT → local LLM via vLLM/Ollama → Piper TTS) achieving ~350ms first-audio / <700ms full-response latency using speculative generation and overlapped sentence TTS, streamed over FastAPI + WebSocket/WebRTC (aiortc) with Silero VAD for turn detection.
- Implemented robust turn-taking and barge-in handling (echo-proofing, prosody + silence + linguistic endpointing), validated on a custom 12-category LLM-judged benchmark scoring 74.5/100, with 0% false endpoints and 0% false backchannel-triggered interrupts (Endpoint P50/P95 = 1.19s / 2.71s).

AI Mentor Chatbot - Medical Exam Prep Assistant

- Built a multi-agent backend using AWS Bedrock, LangChain, and Claude 3 with specialized Query, NCERT, News, and Synthesis agents for exam prep (~95% classification accuracy).
- Deployed a production FastAPI service integrated with DynamoDB, Redis, PostgreSQL, and Docker, incorporating NCERT content with citation tracking and CloudWatch monitoring.

TECHNICAL SKILLS

Languages	Python, Rust (Beginner), SQL
ML / DL	TensorFlow, PyTorch, Scikit-learn, OpenCV, Pandas, NumPy, SciPy
NLP & Speech	Transformers, Whisper, Qwen-Audio, Silero VAD, STT/MT/TTS pipelines.
Gen AI & RAG	LangChain, LangGraph, OpenAI API, RAG pipelines, Agentica A, multi-agent orchestration, LLM fine-tuning.
Hardware / Infra	Qualcomm Cloud AI 100 (QAIC), Docker, AWS (S3, Bedrock, ECS, CloudWatch), CI/CD
Model Inference & Optimization	vLLM, Ollama, CTranslate2, ONNX Runtime, ONNX QNN, QEFFAutoModelForSpeechSeq2Seq, model quantization (INT8/FP16), speculative decoding, KV-cache optimization, batched/streaming inference, on-device NPU inference, edge deployment
Databases	MySQL, SQLAlchemy, DynamoDB, MongoDB, Redis, Pinecone
Web & APIs	FastAPI, REST API, React.js
Tooling	Git, GitLab, Jupyter, Google Colab, Figma, Tableau, AWS QuickSight.

PATENTS & PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T= THESIS

[J.1] Automated Bot Detection on Twitter Using URL Patterns and Learning Automata - Int. J. Eng. Tech. & Mgmt. Sci., Vol. 8, Issue 3, May–Jun 2024. DOI: 10.46647/ijetms.2024.v08i03.025. *PUBLISHED*

[C.1] Speaking Hands: Advancing Communication with Gesture Recognition Systems - ICCICN-24 (Int. Conference on Computational Intelligence & Communication Networks) *PUBLISHED*

[P.1] Real-Time Violence Detection System Using AI for CCTV Surveillance - Patent Application No. 202341029626. ML algorithm detecting violence in real-time CCTV streams with immediate alert generation.

[S.1] DialectSignal: Real-Time Bilingual Arabic–English Speech-to-Speech for Gulf Enterprise - *In preparation; targeting ArabicNLP 2026 (ACL) / IEEE SLT 2026 / IEEE Access.*

OPENSOURCE CONTRIBUTIONS

Semantica × CrewAI Integration · feat/crewai-integration
github.com/semantica-agi/semantica

- Designed and implemented a first-class CrewAI integration for Semantica (semantica[crewai] extra) - SemanticaKGTool, SemanticaDecisionTool, and SemanticaKnowledgeSource as drop-in BaseTool / BaseKnowledgeSource components with 5 KG and 5 decision-intelligence actions, checkpoint-safe serialization, and graceful degradation when CrewAI is absent.
- Hardened via code review (per-graph locking, undirected traversal, type-aware policy coercion), documented the chromadb CVE-2026-45829 conflict, and delivered 70+ tests across an 11-commit series (~71 files changed, 16.2k insertions).

LEADERSHIP & COMMUNITY

Student Placement Coordinator · IIIT Hyderabad *May 2025 - July 2026*

- Coordinated end-to-end campus recruitment for 50+ companies; liaised between recruiters, HR teams, and students; engaged 50+ CEOs and senior business leaders.

Student Dean - IIIC & Sports · SSG-HITAM *May 2023 - May 2024*

- Bridged academia and industry through IIIC; promoted teamwork via institute-level sports events.

Head of R&D · Google Developer Student Club, HITAM *Aug 2022 - Aug 2023*

- Led R&D initiatives and guided teams on project execution; drove innovation across multiple student-run technology projects.

CERTIFICATIONS, AWARDS & HONORS

HydPy NVIDIA Nemotron™ 3 Super Workshop - Top 3 Winner | **KAVACH 2023 Cyber Hackathon - 2nd Place** | **Student of the Department Award - CSE 2024 (HITAM)** | **Hugging Face Agents Course** - Fundamentals of Agents (Sep 2025) |**NPTEL - Python & Deep Learning** (IIT Kharagpur, 83%& 50%) |**Parallel Programming** - OpenMP & CUDA